

MTH 377/577 Convex Optimization

Mid-semester 2025-26

Instructions: Answer all questions. Time: 1 hour.

1. Answer any one of the following:

- (a) Show whether the set $A = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n \mid x_i x_j < 1, \forall i, j = 1, 2, \dots, n\}, i \neq j$ is (i) closed; (ii) bounded. (3)

Ans. (i) Not closed. Consider the sequence,

$$x^{(k)} = \left(1 - \frac{1}{k}, 1 - \frac{1}{k}, \dots, 1 - \frac{1}{k}\right).$$

Then for all i, j , we have $x_i^{(k)} x_j^{(k)} = \left(1 - \frac{1}{k}\right)^2 < 1$, so $x^{(k)} \in A$ for all k . However, $x^{(k)} \rightarrow (1, 1, \dots, 1)$ as $k \rightarrow \infty$ but $(1, 1, \dots, 1) \notin A$. Hence, A is not closed. [Other similar counter examples will also be considered.]

(ii) Not Bounded. Consider $x^{(t)} = (t, 0, 0, \dots, 0)$, $t > 0$. Then for all $i \neq j$, we have $x_i^{(t)} x_j^{(t)} = 0 < 1$, so $x^{(t)} \in A$. However, $\|x^{(t)}\| = t \rightarrow \infty$ as $t \rightarrow \infty$. Hence, A is not bounded. [Explanation: note that since $x^{(t)} \rightarrow \infty$ as $t \rightarrow \infty$, $\nexists M \neq \infty$ such that $\|x^{(t)}\| \leq M$. A is not bounded below.]

- (b) Consider the cone generated by the vectors $\{(1, 2, 0), (2, 1, 0)\}$. Find its dual. Is the dual convex? (3)

Ans. $C = \{\theta x \in \mathbb{R}^3 \mid x \in \{(1, 2, 0), (2, 1, 0)\}, \theta \in \mathbb{R}_+\}$ is the cone generated by the given vectors. (You may also take conic combinations of the two vectors). Dual of C is

$$C^* = \{y \in \mathbb{R}^3 \mid y \cdot x \geq 0 \text{ for all } x \in C\}.$$

for the given cone, the above becomes:

$$C^* = \{(y_1, y_2, y_3) \in \mathbb{R}^3 \mid y_1 + 2y_2 \geq 0, 2y_1 + y_2 \geq 0\}.$$

2. Solve the following maximization problem using KKT conditions. Show all the steps: (5)

$$\begin{aligned} \text{maximize} \quad & 2x_1 + 2x_2 + 2x_3 - x_1^2 - x_2^2 - x_3^2 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 \leq 3, \\ & x_1 \geq 0, \ x_2 \geq 0, \ x_3 \geq 0. \end{aligned}$$

Ans. $x^* = (1, 1, 1)$.

$$f_1(x) = 3 - x_1 - x_2 - x_3 \geq 0$$

$$f_2(x) = x_1 \geq 0, \quad f_3(x) = x_2 \geq 0, \quad f_4(x) = x_3 \geq 0$$

Objective function

$$f_0(x) = 2x_1 + 2x_2 + 2x_3 - x_1^2 - x_2^2 - x_3^2$$

Derivatives of the objective function

$$\nabla f_0 = (2 - 2x_1, 2 - 2x_2, 2 - 2x_3)$$

Derivatives of constraints

$$\nabla f_1 = (-1, -1, -1)$$

$$\nabla f_2 = (1, 0, 0), \quad \nabla f_3 = (0, 1, 0), \quad \nabla f_4 = (0, 0, 1)$$

KKT Condition

$$\nabla f_0 + \lambda_1 \nabla f_1 + \lambda_2 \nabla f_2 + \lambda_3 \nabla f_3 + \lambda_4 \nabla f_4 = 0$$

$$(2 - 2x_1, 2 - 2x_2, 2 - 2x_3) + \lambda_1(-1, -1, -1) + \lambda_2(1, 0, 0) + \lambda_3(0, 1, 0) + \lambda_4(0, 0, 1) = 0$$

Thus

$$2 - 2x_1 - \lambda_1 + \lambda_2 = 0$$

$$2 - 2x_2 - \lambda_1 + \lambda_3 = 0$$

$$2 - 2x_3 - \lambda_1 + \lambda_4 = 0$$

Complementary Slackness

$$\lambda_1(3 - x_1 - x_2 - x_3) = 0$$

$$\lambda_2 x_1 = 0, \quad \lambda_3 x_2 = 0, \quad \lambda_4 x_3 = 0$$

Case 1:

$$\lambda_2 = \lambda_3 = \lambda_4 = 0$$

and $\lambda_1 \neq 0$

Then,

$$\lambda_1 = 2 - 2x_1 = 2 - 2x_2 = 2 - 2x_3$$

$$x_1 = x_2 = x_3$$

using $3 - x_1 - x_2 - x_3 = 0$

$$x_1 + x_2 + x_3 = 3$$

$$3x_1 = 3$$

$$x_1 = 1$$

Hence

$$x_1 = x_2 = x_3 = 1$$

From the above we know that $(1, 1, 1)$ is a KKT point. Case 2: $x_1 = x_2 = x_3 = 0$. Then $\lambda_1(3 - x_1 - x_2 - x_3) = 0$ implies $\lambda_1 = 0$. Then $\lambda_2, \lambda_3, \lambda_4$ may not all be zero. Therefore, $(0, 0, 0)$ is a KKT point. Since $f(1, 1, 1) > f(0, 0, 0)$, the maxima is $x^* = (1, 1, 1)$.

3. Consider the following optimization problem: *maximize* f_0 s.t. $f_i \geq 0$, $i = 1, 2, \dots, m$ where f_0 and each f_i are concave, continuous and differentiable on an open convex set C . Let x^* be a KKT point. Is x^* a maximizer of the problem? Can f_0 have a maximizer that is not a KKT point? (4)

Ans. Yes. KKT is sufficient for optimality for a concave program. Therefore x^* is a maximizer (under strong duality).

If x^* is a KKT point, then there exist $\lambda_p \geq 0$ such that

$$\nabla f_0(x^*) + \sum_p \lambda_p \nabla f_p(x^*) = 0, \quad \lambda_p f_p(x^*) = 0.$$

Hence x^* maximizes $F(x) = f_0(x) + \sum_p \lambda_p f_p(x)$, which implies $f_0(x^*) \geq f_0(x)$ for any feasible x .

Yes, f_0 can have a maximizer that is not a KKT point. For example: let $f_0(x) = x$ and $f_1(x) = -x^2$, $x \in \mathbb{R}$. The feasible set is $\{0\}$, so $x^* = 0$ is the maximizer. Now, $\nabla f_0(0) = 1$, $\nabla f_1(0) = 0$. The KKT condition requires: $\nabla f_0(0) + \lambda \nabla f_1(0) = 0$, which gives $1 + \lambda \cdot 0 = 0$, which is impossible. Therefore, there is no λ that satisfies KKT condition.

4. Let $f(x_1, x_2) = x_1^3$ and $g(x_1, x_2) = x_2^3$ for $(x_1, x_2) \in \mathbb{R}^2$. Are f and g quasiconvex? What about $f + g$? (4)

Ans. f and g are quasiconvex since their domain and all sublevel sets are convex [Lecture 9]. However, $h = f + g$ is not quasiconvex. Consider the points $x = (-2, 1)$, $y = (1, -2)$, $\theta = \frac{1}{2}$. Then, $h(x) = (-2)^3 + 1^3 = -8 + 1 = -7$, $h(y) = 1^3 + (-2)^3 = 1 - 8 = -7$. So, $\max\{h(x), h(y)\} = -7$. Also, $\theta x + (1 - \theta)y = (-0.5, -0.5)$,

$$\implies h(-0.5, -0.5) = (-0.5)^3 + (-0.5)^3 = -0.125 - 0.125 = -0.25.$$

Thus, $h(\theta x + (1 - \theta)y) = -0.25 > -7 = \max\{h(x), h(y)\}$, violating the quasiconvexity condition.

5. Let A, B be non-empty, convex, disjoint, closed sets in $\mathbb{R}^n$. Does there exist a strict separating hyperplane between any such A and B ? Provide a short proof/counterexample. (4)

Ans. No, a strict separating hyperplane does not necessarily exist. The sufficiency condition for the existence of a hyperplane between A, B requires boundedness. For a counterexample: see the lecture slide. There may exist some sets A and B , but the question states “any such A and B ”, so counterexamples proving that a strict separating hyperplane exists, isn’t valid.